Cape Peninsula University of Technology

Subject Name: *Project 2*

Subject Code: *PRT262S*

Supervisor: Mr. R.Burger

Group: R Burger 4

Group Members:

Anelisiwe Cele - 230236901

Khanya Cezula - 231323573

Amukelani Maurel Chauke - 250055392

Beyonce Cloete - 240822757

Zakariyya Cloete- 230939023

Thabiet Darries - 221106901

Lost and Found System

/ Campus Finder *

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Problem Definition

Students often lose personal belongings on campus such as phones, student cards, books, and other valuable items. Currently, there is no centralized platform that allows students to easily report lost items or check if their items have been found. This makes the process of recovering lost belongings slow and inefficient.

The proposed Lost and Found System will provide a digital platform where students can report lost or found items, upload images, and search for items based on category, location, or date. This system will make it easier for students to locate their belongings and improve the overall management of lost property on campus.

Project Scope

Project Name : **Lost and Found System/Campus Finder***	
Project Manager : Zakariyya Cloete	
Project Deadline : TBD	
Project Goal :	
This project aims to help users who have lost personal items on campus make a report to find their items in a quick and swift manner.	
Project Budget :	
No Budget Allocated	
Project Deliverables :	
Deliverables	Description
System Requirements Documentation	A document outlining the functional and non-functional requirements of the Lost and Found System.
Lost and Found System Prototype	A working prototype of the system that allows users to register, log in, report lost items, report found items, and search for items.
Testing Documentation	Test cases and results demonstrate that the system functions correctly.

Final Project Documentation	A report describing the system, design decisions, and how the project was completed.

Project Objectives :

Develop a user-friendly system for reporting lost and found items

- Ensure that students, staff, and campus personnel can easily register, log in, and submit reports for lost or found items.

Enable quick and efficient item tracking and retrieval

- Implement search and claim features to help users locate their lost belongings in a swift manner.

Ensure accurate recording of user and item information

- Maintain a reliable database that stores item details, ownership information, and timestamps for reporting and claiming.

Test and validate system functionality

- Conduct thorough testing to ensure all features work correctly and the system meets its functional and non-functional requirements.

Provide comprehensive project documentation

- Deliver clear reports, including system requirements, design decisions, testing outcomes, and final project documentation to support maintenance and future improvements.

Out Of Scope : (Reasoning**)**

The Project will not include the following

- Real-Time tracking locations - Not included because it does not require live tracking.
- Payment Systems – We are only focusing on reporting and retrieving lost items.
- AI Assistance – Beyond our current skill level.
- Mobile Application – We are currently focusing on developing a Desktop application first.

Technologies Used :

Programming Language

The Lost and Found System will be developed using Java as the main programming language. Java is an object-oriented programming language that allows the system to be structured using reusable classes such as users, items, and posts. This approach allows to have a maintainable and structured codebase.

Database Technology

As a group we're still deciding on how to store data and what database to use. We are unsure about using MySQL as the database management system or Java Database Connectivity (JDBC) to manage database interaction with the application. A choice will be made during the development phase of the project.

Integrated Development Environment (IDE)

The development of the system will take place in NetBeans. NetBeans provides us tools such as debugging support, and code completion. Which can help save time and improve efficiency.

Execution Platform

The Lost and Found System will operate as a window-based desktop application. The system will run on computers on the campus and allow users to report lost items, submit found items, and search the system for their personal items.

Functional Requirements

FR-ID	Functional Requirement	Priority
FR0001	Users must be able to register and log into the system.	High
FR0002	Users must be able to report lost or found items.	High
FR0002	Users must be able to upload images of items.	Medium
FR0004	Users must be able to search and filter items by category, date, or location.	High
FR0005	The system must allow items to be marked as claimed or returned.	High
FR0006	Users must be able to view detailed information about an item.	Medium
FR0007	Users must be able to edit or delete their item posts.	High

Non-Functional Requirements

NFR-ID	Non-Functional Requirement	Description
NFR0001	Usability	The system must have a simple and user-friendly interface so that students can easily report and search for items.
NFR0002	Security	The system must protect user accounts and personal information through secure login and authentication.
NFR0003	Performance	The system must respond quickly when users search for items, report items, or update item information.
NFR0004	Reliability	The system should function consistently without frequent errors or crashes.
NFR0005	Compatibility	The system should run on common devices and browsers used by students.
NFR0006	Data Integrity	The system must ensure that item information and user data are stored accurately in the database.

Users Benefiting from the Lost and Found System

Students

Benefit: Students can report lost items or claim items they have found on campus.

Role in the system:

- Register and log in to the system
- Report lost items
- Submit details of found items
- Search the system for their lost belongings
- Claim items that belong to them

Staff / Lecturers

Benefit: Staff members can also recover lost items or report items found in classrooms, offices, or campus areas.

Role in the system:

- Register and log in to the system
- Report lost items
- Submit details of found items
- Verify ownership when claiming an item
- Provide location details of where items were found

Cleaning Staff

Benefit: Allows staff who find lost items during cleaning or maintenance to easily report them.

Role in the system:

- Submit found items to the system
- Provide location details of where items were found

Group Organization

Member	Role	Responsibilities	Reason for Allocation
Zakariyya Cloete	Project Manager / Chairperson	Leads the group, organizes meetings, monitors progress, and ensures project deadlines are met.	Selected due to strong leadership, organization, and time-management skills.
Beyonce Cloete	Technical Lead / Secretary	Handles technical aspects of the project and records meeting minutes. Responsible for maintaining project documentation.	Chosen because of strong technical knowledge and good organizational abilities.
Khanya Cezula	Business Analyst	Identifies the problem, gathers system requirements, and ensures the system meets user needs.	Selected due to strong analytical thinking and research skills.

Amukelani Maurel Chauke	System Designer	Designs the overall system structure and plans how system components will interact.	Allocated this role because of logical thinking and system planning skills.
Thabiet Darries	UI/UX Designer	Designs the user interface and ensures the system is user-friendly and visually appealing.	Chosen because of creativity and interest in design.
Anelisiwe Cele	Database Designer	Designs and manages the database structure and ensures data is stored and organized correctly.	Selected due to attention to detail and ability to manage and organize data.

Task Allocation

Task Allocation to Group Members

Member	Role	Assigned Tasks
Zakariyya Cloete	Project Manager	Organize meetings, track progress, ensure deadlines are met
Beyonce Cloete	Scribe / Technical Lead	Record meeting minutes and assist with report preparation
Khanya Cezula	Business Analyst	Define system requirements, assist with documentation
Amukelani Maurel Chauke	System Designer	Design system structure and diagrams
Thabiet Darries	UI/UX Designer	Design interface layout and ensure usability
Anelisiwe Cele	Database Designer	Design database structure and manage data organization

Skills Contributed by Each Member

Member	Skills Contributed
Zakariyya Cloete	Leadership, organization, project coordination, and time management.
Beyonce Cloete	Technical knowledge, problem solving, documentation, and meeting organization.
Khanya Cezula	Analytical thinking, research skills, and requirement documentation.
Amukelani Maureal Chauke	System planning, logical thinking, and design skills.
Thabiet Darries	Creativity, interface design, and user experience understanding.
Anelisiwe Cele	Data organization, database design, and data management skills.

Provisional Project Timeframes

Project Phase	Time Frame
Project planning and requirement analysis	Feb – March
System design and architecture planning	March – May
System development and implementation	May – September
System testing and improvements	September – November
Final documentation and project completion	November – December

Time Frames for Learning New Skills

Skill	Learning Time Frame
Improving Java programming skills	March – May
UI/UX design tools and interface development	May – August
Learning database design and SQL	August – September
System testing and debugging techniques	September - December

Communication

Communication Methods Used

Our primary communication method was WhatsApp because the App is used in our everyday life. By using WhatsApp, we can see notifications, updates, respond instantly and discuss project tasks. During meetings(Video Calls) we can share our screens, making sharing information and presentations easier.

What Worked Well

- All members participated and Contributed to discussions
- Effective Communication
- Video Calls worked effectively by doing presentations online

What Was Tried but Less Effective

- Blackboard was our initial approach but due to blackboards delayed push notifications. This made it difficult for members to get notifications on time. As a result, we decided to use a different communication method therefore WhatsApp was adopted as our primary communication platform.

Communication Challenges

- Poor or a lack of internet connection can prevent members not to receiving important updates or attend online meetings. During one of the meetings, a group member experienced network issues. This resulted to the member unable to join the online session and miss out on important updates.

Communication Per Member

Member	Communication	Activity in Meetings
Zakariyya Cloete	Active	Leads discussions and coordinates meeting progress
Beyonce Cloete	Limited	Mainly listens and records important meeting notes
Khanya Cezula	Active	Participates when providing input or clarification
Amukelani Maureal Chauke	Active	Listen and contribute when necessary
Thabiet Darries	Active	Participates in technical discussions
Anelisiwe Cele	Active	Contributes ideas and participates in discussions

MEETING AGENDA

PTA

DATE:	TIME:	MEETING LOCATION:
26/02/26	11:20 AM	WhatsApp Group Call

MEETING CALLED BY:	Zakariyya Cloete
TYPE OF MEETING:	Discussion
FACILATATOR:	Zakariyya Cloete
NOTE TAKER:	Beyonce Cloete
TIMEKEEPER:	Beyonce Cloete

IN ATTENDENCE:	APOLOGIES:
Zakariyya Cloete	Everyone was present
Beyonce Cloete	
Khanya Cezula	
Amukelani Cele	
Thabiet Darries	
Anelisiwe Cele	

TOPICS

TIME ALLOATED	AGENDA TOPIC:	PRESENTER:
20 min	Core Functional Requirements	Zakariyya Cloete

ACTION ITEMS	PESRON RESPONSIBLE	DESCRIPTION
Implement Login/Signup	Zakariyya Cloete	Create user authentication for students
Add category filter	Zakariyya Cloete	Filter items by tech, student materials, etc.
Develop "lost item" feature	Zakariyya Cloete	Allow users to submit lost item reports
Develop "found item" feature	Zakariyya Cloete	Allow users to report found items
Implement picture upload/ description post	Zakariyya Cloete	Users can upload images or post descriptions
Standardize description fields	Zakariyya Cloete	Lost items: lost location, last seen, item description. Found items: location found, date found, item details

TIME ALLOTED:	AGENDA TOPIC:	PRESENTER:
20 min	Security & Verification Features	Thabiet Darries

ACTION ITEMS	PERSON RESPONSIBLE	DESCRIPTION
Add verification questions	Amukelani Chauke	Security questions before claiming items
Phone claim security rule	Thabiet Darries	Description required (make, model, year) instead of image upload
Ownership auto-verification rule	Thabiet Darries	If an ID/student card is found, the verification process is reduced since the owner is identifiable

TIME ALLOTED:	AGENDA TOPIC:	PRESENTER:
20 min	Community & Platform Scope	Khanya Cezula

ACTION ITEMS	PERSON RESPONSIBLE	DESCRIPTION
Add comment section	Khanya Cezula	Allow users to comment on posts
Add community group chat	Khanya Cezula	Campus-wide discussion/chat feature
Restrict application scope	Khanya Cezula	App is campus-based only and does not include items lost/found outside of campus

TIME ALLOCATED:	AGENDA TOPIC:	PRESENTER:
20 min	Optional system/Backend enhancements	Anelisiwe Cele

ACTION ITEMS	PERSON RESPONSIBLE	DESCRIPTION
User role management	Anelisiwe Cele	Admin vs student access control
Moderation system	Khanya Cezula	Preventing spam or false claims

Next meeting: 12/03/26 14:15

Agenda: System Flow Diagram

MEETING AGENDA

PTA

DATE:	TIME:	MEETING LOCATION:
12/03/26	14:15	WhatsApp Group Call

MEETING CALLED BY:	Zakariyya Cloete
TYPE OF MEETING:	Discussion
FACILIATTOR	Zakariyya Cloete
NOTE TAKER:	Beyonce Cloete
TIMEKEEPER:	Beyonce Cloete

IN ATTENDENCE:	APOLOGIES
Zakariyya Cloete	Everyone was present
Beyonce Cloete	
Khanya Cezula	
Amukelani Chauke	
Anelisiwe Cele	
Thabiet Darries	

TOPICS

TIME ALLOTED:	AGENDA TOPIC:	PRESENTER:
30 min	System Flow Diagram	Zakariyya Cloete

Action items	Person responsible	Description
Add functionality	Thabiet Darries	Implement other features that are not present in the current diagram
Modify diagrams	Amukelani Chauke	Update flow diagrams throughout the project duration

TIME ALLOCATED:	AGENDA TOPIC:	PRESENTER:
30 min	Member Roles	Anelisiwe Cele

Action items	Person responsible	Description
Project Manager & Team Leader	Zakariyya Cloete	Schedules meetings and communicates with the supervisor
Technical Lead & Secretary	Beyonce Cloete	Coder. Oversee development and make technical decisions
Business Analyst	Khanya Cezula	Defines system requirements
System Designer	Amukelani Chauke	Designs overall system structure
UI/UX Designer	Thabiet Darries	Designs interface layout and ensure system is user friendly.

Database designer	Anelisiwe Cele	Designs the database

Next meeting: TBD

Agenda: TBD